ICTD COURSE CASE STUDY

Commons Connect: A Digital Participatory Tool for Water Security

Mapping an ICTD research project onto the six topics of Information and Communication Technologies for Development and Sustainability

Based on: Mehta, Dutt, et al. "Initial Observations from Field Testing of a Digital Participatory Tool to Improve Water Security in Rural India." ICTD 2024, Nairobi, Kenya.

Course: ICTD — Instructor Aaditeshwar Seth, IIT Delhi • Deck 1 of 2 (Appendix deck covers Impact Evaluation research-question detail)

July 2026

WHERE WE ARE GOING

Roadmap: Six Topics, One Paper

1	**Problem Discovery**	*Section 3 — formative research across four blocks*
2	**Computing Fundamentals**	*Section 4 — the CoRE Stack: datasets, indicators, tools*
3	**Ethnographic Design & Iteration**	*Section 5 — field testing, usability, feature requests*
4	**Socio-technical Dynamics at Scale**	*Section 5.3–5.4 — equity mapping, stakeholder politics*
5	**Impact Evaluation**	*A companion RCT concept note (2026) — not yet in the paper*
6	**Operations & Scaling**	*Section 4.4 + a 2026 field update from Ramgarh, Jharkhand*

PART 1 OF 6

Problem Discovery

Deeply understanding context, constraints, incentives, and existing systems before assuming you know the problem.

Maps to: Commons Connect paper, Section 3 — Formative Research

CONTEXT

MGNREGA: The Institutional Backdrop

- MGNREGA (2005): guarantees 100 days of wage employment/year for rural households; labour is meant to build durable natural-resource assets.
- A large share of the budget targets water security — farm ponds, check-dams, trenches, bunds, well repair — for irrigation and climate resilience.
- Demand-driven, in principle: Gram Sabha (village meeting) → Panchayat aggregation/review → Block-level allocation.
- Civil Society Organizations (CSOs) frequently support this process on the ground — helping communities identify needs and navigate the bureaucracy.
- **Prior GIS efforts (Bhuvan, CRISP-M, Yuktdhara, CLART) exist but are coarse, staff-only, or disconnected from bottom-up community input — the gap Commons Connect is designed to fill.**

Formative Research: Four Blocks, One Method

- Identical field protocol replicated across four administrative blocks, chosen for climatic and land-use diversity:
    - Angul (Odisha) — undulating, tribal/forest communities
    - Masalia (Jharkhand) — hills/plateaus, hard-rock aquifer
    - Mohanpur (Bihar) — Indo-Gangetic plain, alluvium aquifer
    - Pindwara (Rajasthan) — Aravalli hills, low rainfall (600–800mm)
- Each site: an agro-ecological expert, a hydrologist, a developer, and a design researcher — a genuinely interdisciplinary team.
- Consultations (15–30 participants each) deliberately spanned daily wage workers, marginal farmers, landless labourers, tribal communities, women's SHGs, CSO staff, and officials.

FINDING 1

No Shared Understanding of the Landscape

- Example (Mohanpur, Bihar): double/triple-cropping areas relied on borewells; groundwater dropped, borewells now reach 300+ feet.
- Open dug wells traditionally used by smaller, rain-fed farmers dried up — Rabi (winter) cropping of wheat, pulses, vegetables became impossible for them.
- **Communities were well aware of the decline individually — but showed no collective response: no self-imposed limits on new borewells, no sharing norms.**
- A classic tragedy-of-the-commons pattern for groundwater — awareness without coordination.
- Design implication: data-driven visualizations of rainfall, drought, groundwater, and cropping change could give CSOs and volunteers a common reference point to anchor these conversations.

FINDING 2

Unscientific Site Selection

- Pindwara (Rajasthan): a check-dam built too far upstream with a tiny catchment — never accumulated enough water, and reduced downstream flow for no benefit.
- Mohanpur (Bihar): isolated farm ponds on flat terrain with no inlets — negligible runoff capture.
- Angul (Odisha): wells sited in low-discharge zones — delayed water accumulation by 8–10 days.
- Masalia (Jharkhand): a well construction had to be abandoned after hitting hard rock.
- **Common thread: site selection driven by targets and convenience, not terrain, soil, or hydrology — the opposite of the ridge-to-valley approach CSOs like FES otherwise advocate.**

FINDING 3

Inequities in Resource Allocation

- Exclusion followed multiple, often compounding, lines: caste (Mohanpur), upstream vs. downstream farmers (Angul, Masalia), political connectedness (Mohanpur), and wealth (Pindwara).
- Downstream, politically-connected, prosperous farmers disproportionately captured allocations for assets like farm ponds.
- Upstream works (e.g., check-dams) require downstream consent — a further leverage point for elites.
- MGNREGA's own mechanics compound this: delayed wage payments and delayed capital-expense reimbursements mean only households with working capital can absorb the float — a structural bias toward the already-prosperous.
- Field staff were often reluctant to intervene, since they depend on Panchayat elites to identify sites and meet top-down targets.

From Findings to Design Goals

- **The three field findings map almost directly onto the three design goals of Commons Connect:**
  - → No shared landscape understanding → build a data-driven shared understanding (visualizations of rainfall, groundwater, cropping change)
  - → Unscientific site selection → scientific, evidence-based site suitability (recharge potential, drainage lines, terrain)
  - → Inequities in allocation → legibility of resource ownership/usage across settlements to inform fairer allocation
- This is the core methodological move worth noticing: the tool's feature set is not a generic GIS wish-list — it is traceable, item by item, to specific field observations.

CoRE stack: Theory of change

Ideal state: Managed agriculture risk, Sustainable ecosystem, Removal of factors of oppression

Bottom-up planning for supply-side allocation

AGRICULTURE

A monumental waste: Crores spent on water-related works haven't readied Jharkhand for droughts; here's why

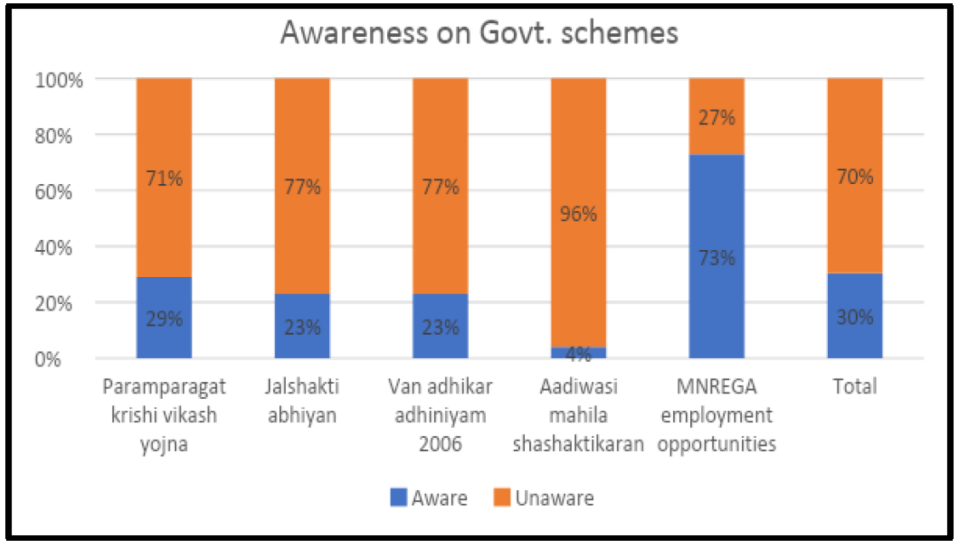
Down to Earth, 2023

CoRE stack: Theory of change

Ideal state: Managed agriculture risk, Sustainable ecosystem, Removal of factors of oppression

Bottom-up planning for supply-side allocation

Informed demand generation



Gram Vaani, 2022

Field visits (Gaya, Sirohi), 2023

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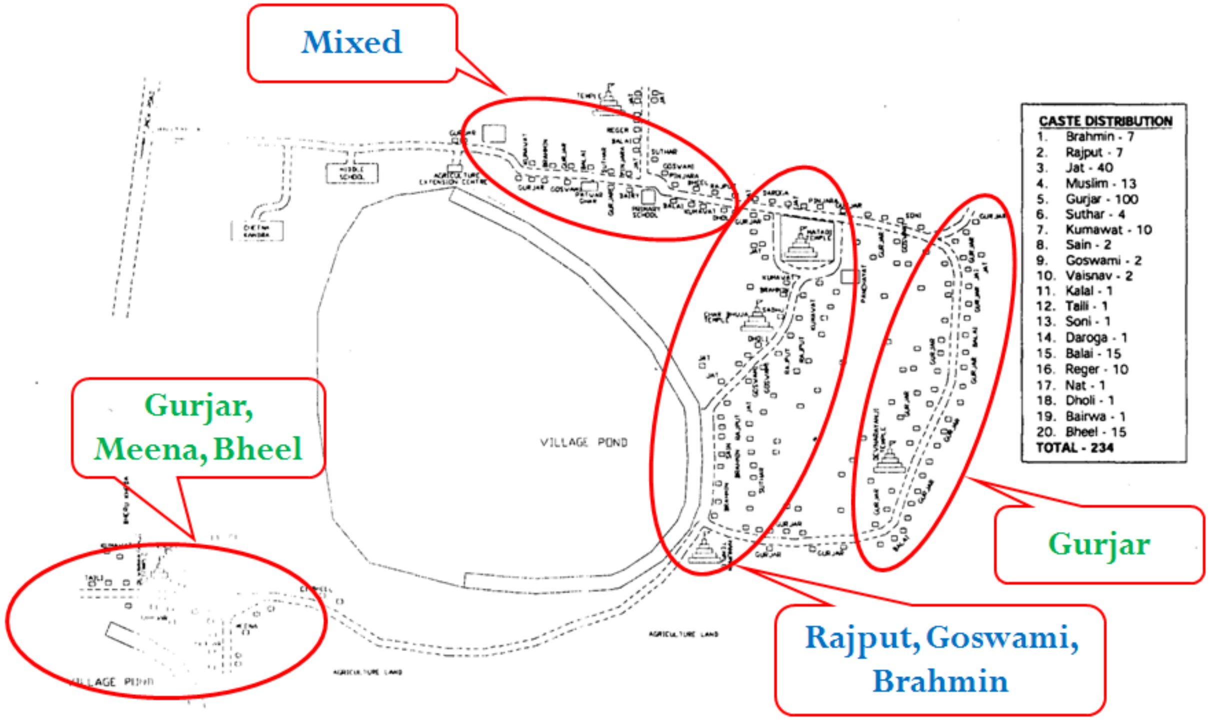
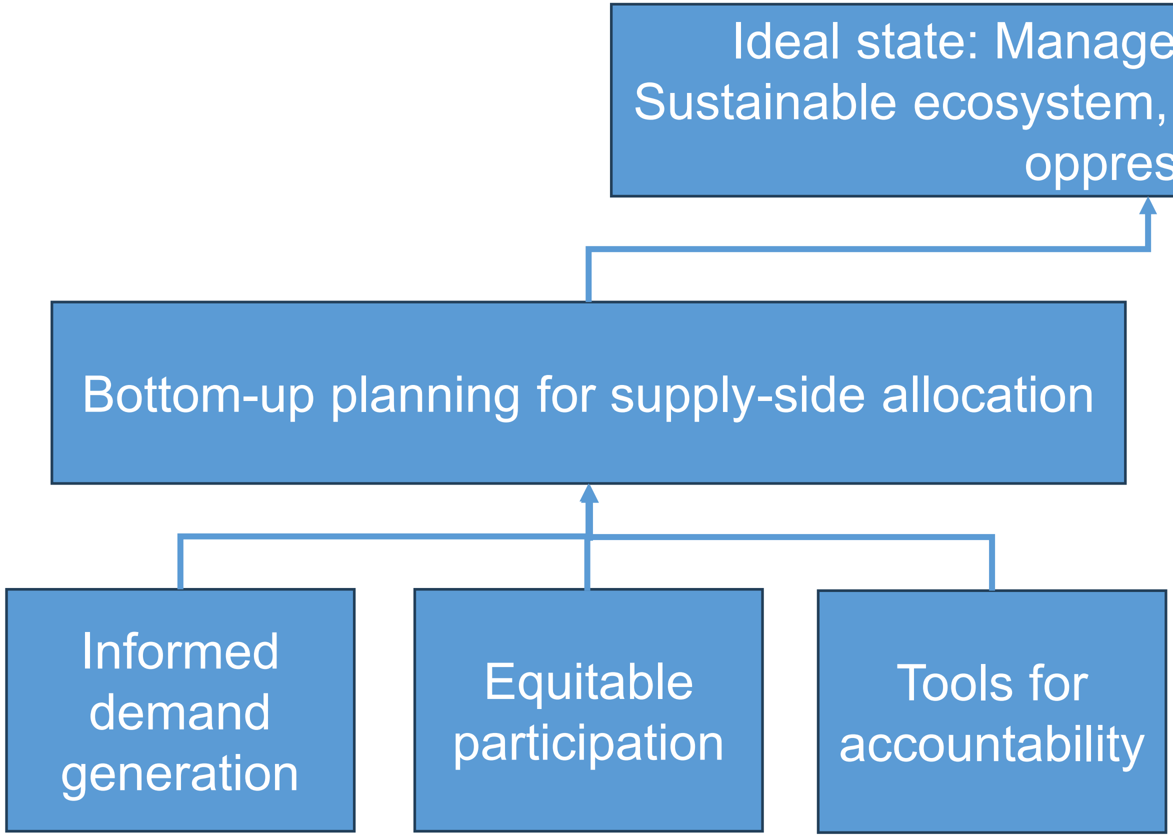
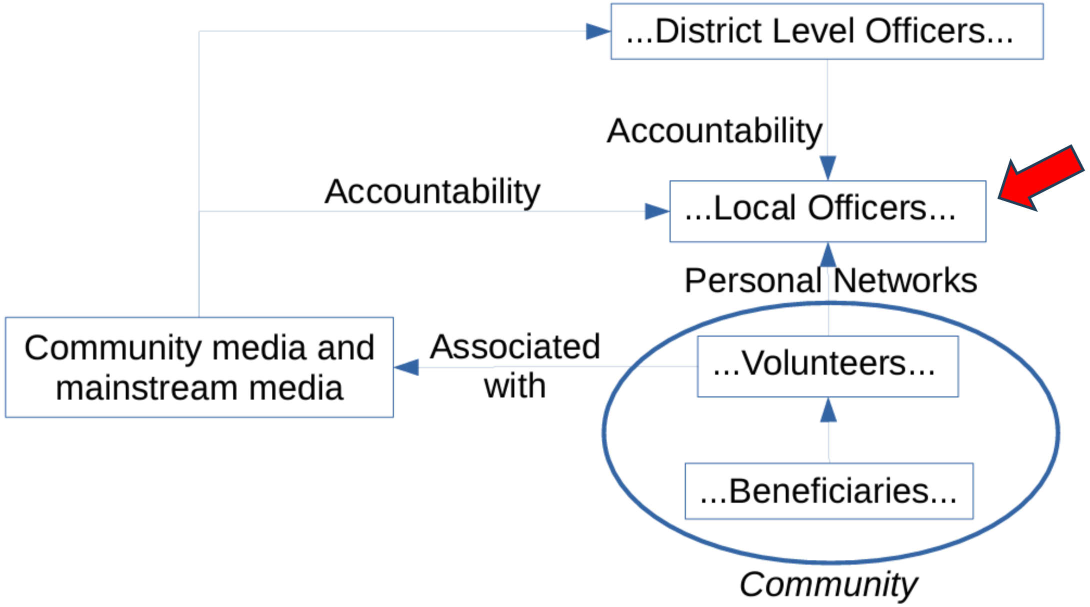
Equitable participation



Krishna, 2002

CoRE stack: Theory of change





Chakraborty, 2017

CoRE stack: Theory of change

Ideal state: Managed agriculture risk, Sustainable ecosystem, Removal of factors of oppression

Bottom-up planning for supply-side allocation

Collectivization for demand-side changes

Informed demand generation

Equitable participation

Tools for accountability

Field visits (Jamui), 2022

CoRE stack: Theory of change

Ideal state: Managed agriculture risk, Sustainable ecosystem, Removal of factors of oppression

Collectivization for demand-side changes

Shared understanding

Field visits (Sirohi), 2023

CoRE stack: Theory of change

Ideal state: Managed agriculture risk, Sustainable ecosystem, Removal of factors of oppression

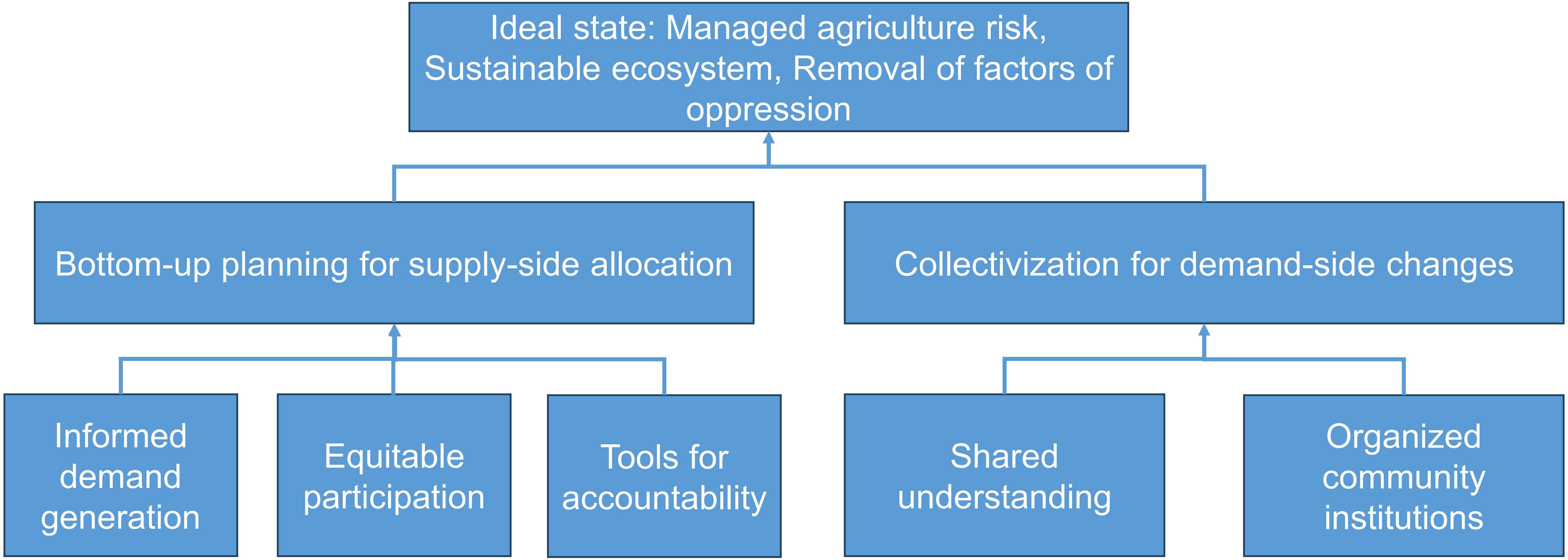
Collectivization for demand-side changes

Shared understanding

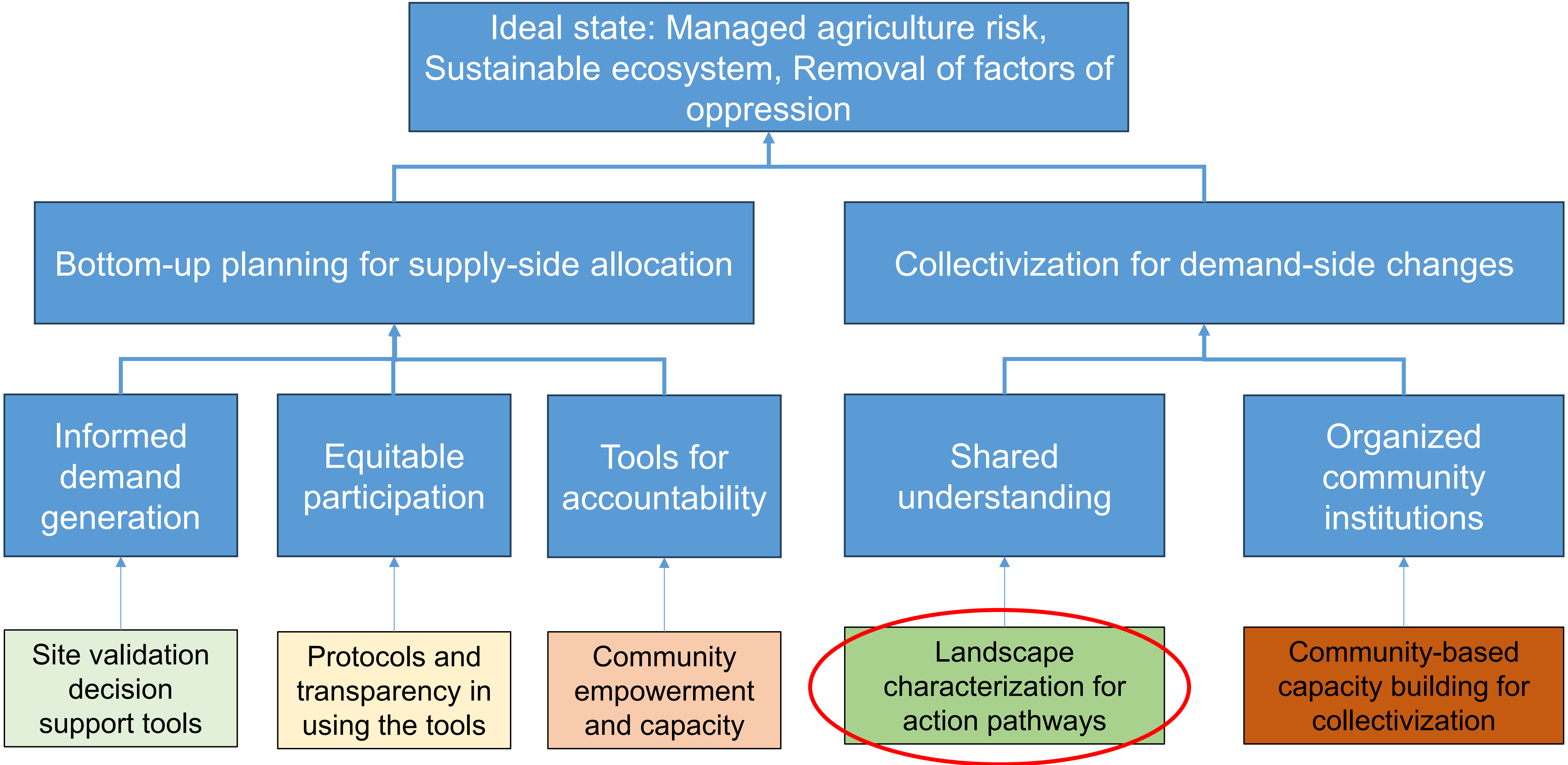
Organized community institutions

Field visits (Latehar), 2023

CoRE stack: Theory of change



CoRE stack: Theory of change



CoRE stack: Theory of change

With a three-pronged approach of:

(a) identifying and training community volunteers to map vulnerable social groups in their community and the allocation of NRM assets to them,

(b) using participatory decision support tools to recommend to authorities useful NRM initiatives, and to communities more responsible cropping patterns, and

(c) manage a local participatory media platform on which relevant information about environmental aspects is discussed,

the proposed intervention model will empower communities to demand social welfare schemes and participate in decision-making processes to strengthen their resilience against water crises, and ensure that the schemes are equitably provisioned and transparently implemented, thereby leading to more sustainable living.

PART 2 OF 6

Solving with Computing Fundamentals

Applying data science, ML, remote sensing, and systems work — built to withstand real constraints of connectivity, devices, and literacy.

Maps to: Commons Connect paper, Section 4 — Commons Connect / the CoRE Stack

ARCHITECTURE

The CoRE Stack: Four Layers

Datasets	ML on satellite imagery + secondary sources: cropping intensity, surface water, land-use, rainfall, runoff, terrain, geology, soil, aquifer maps.
Indicators	Datasets aggregated to village/panchayat or (micro-)watershed units: groundwater change, drought incidence, cropping-drought sensitivity, water stress.
Tools	Commons Connect: visualizations, site-assessment, equity/resource mapping, community demand logging. Other tools (agroforestry, forest restoration, carbon tracking) can plug into the same stack.
Gov't Integration	Auto-generated Detailed Project Report (DPR) for submission into the MGNREGA workflow; API integration with platforms like Yuktdhara where needed.

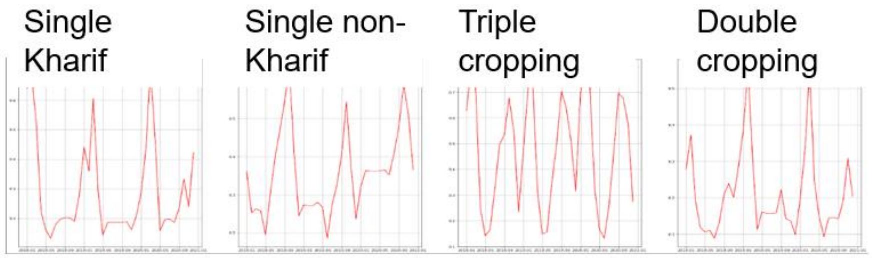
Layers build on each other: raw pixels become indicators become interactive tools become government-recognized planning documents.

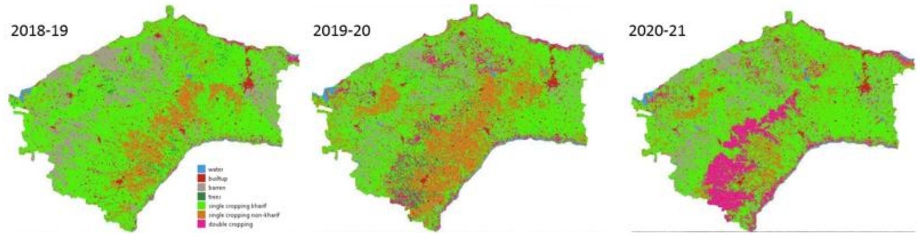
Why the Micro-Watershed?

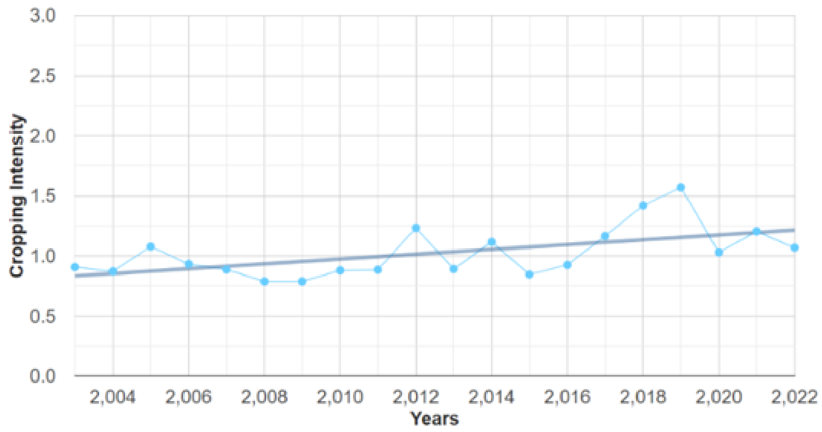
- Block (~100 sq. km, ~a dozen Panchayats) is the level at which MGNREGA resource-allocation decisions are actually made — so it's the outer boundary for data generation.
- Within a block, Commons Connect works at the micro-watershed: a hydrological unit with a single rainfall-runoff outlet, ~1000 hectares (roughly 3km × 3km).
- Delineated using ridgeline-identification algorithms on digital elevation maps, starting from India-WRIS river-basin boundaries.
- **1000 ha is deliberately chosen to match the areal extent of a typical Indian village — small enough for communities to recognize and relate to on a map, not an abstract administrative polygon.**
- Trade-off acknowledged in the paper: this framing captures vertical flux well but omits incoming runoff from upstream micro-watersheds and lateral groundwater flow — a real modeling simplification.

Intra-annual Land Use / Land Cover (LULC)

- Problem with existing products: most global LULC products (e.g. Dynamic World) are annual and cross-sectional — they miss within-year seasonal dynamics that matter for irrigation planning.
- India has three cropping seasons: Kharif (monsoon, largely rain-fed), Rabi (post-monsoon, irrigation-dependent), Zaid (summer, almost entirely groundwater-dependent).
- Commons Connect's contribution: a custom intra-annual classifier using time-series data across all seasons, at 10m × 10m pixel resolution, trained on samples from all of India's agro-climatic zones.
- **Reported accuracy: 83% for cropping-intensity classification; 94% for standard LULC classes (water, trees, crops, etc.).**
- Runs on Google Earth Engine, producing annual outputs from 2003 to present for any area of interest.



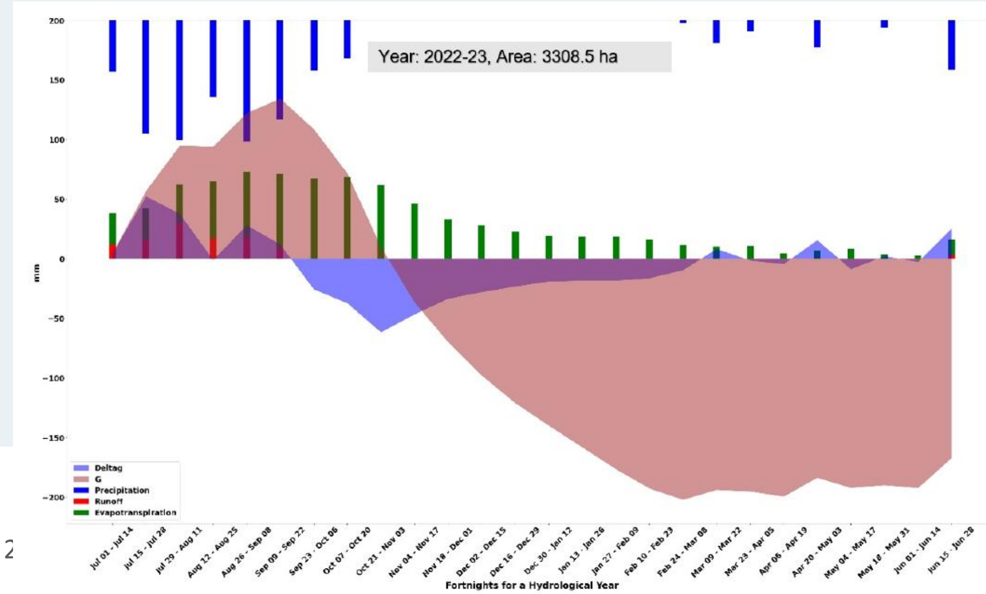




Hydrological Inputs: Rainfall, ET, Runoff

Precipitation & Evapotranspiration

- Rainfall: JAXA global product downloaded and processed into fortnightly micro-watershed values.
- Evapotranspiration (ET): daily FLDAS data (validated in tropical settings) aggregated to fortnightly ET per micro-watershed.



Runoff

- No transparent government product exists at seasonal resolution (Bhuvan's runoff maps are annual and undocumented).
- Commons Connect implements a pixel-by-pixel runoff pipeline using daily rainfall, Hydrologic Soil Group, slope, and land-use — aggregated fortnightly per micro-watershed.
- Planned extension: incorporating incoming runoff from upstream micro-watersheds, with field validation still pending.

From Datasets to Groundwater Change

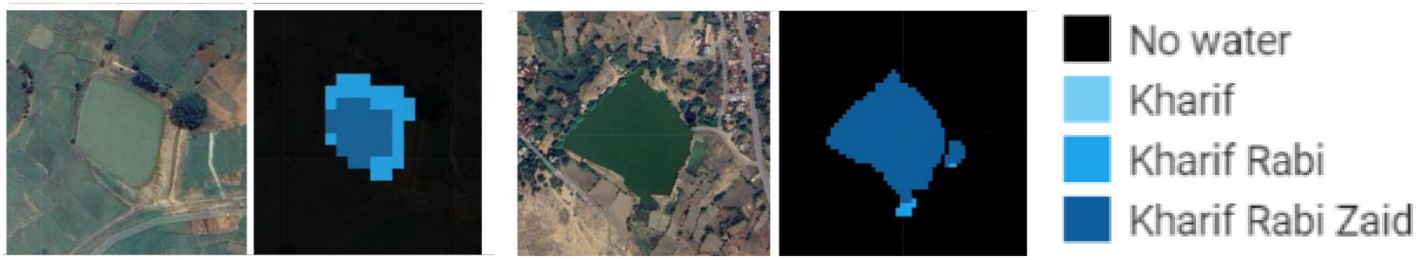
- Method: water-balance approach — precipitation is split between outgoing runoff, evapotranspiration, and the unknown, which is solved for as vertical groundwater flux.

```
P  =  Q(out)  +  ET  +  ΔG(unknown)
```

- ΔG (change in groundwater volume) is then converted to an observable change in well depth using the aquifer's specific yield (from CGWB's aquifer mapping, via WRIS).
- **Calibration issue found in field testing: the model matched observations well in Masalia (crystalline aquifer, dominated by vertical flux) but not in Mohanpur (alluvium aquifer, heavy lateral flow) — a genuine limitation of the vertical-flux-only assumption.**
- Ongoing work: aquifer-type-specific calibration models, using CGWB well data as ground truth.

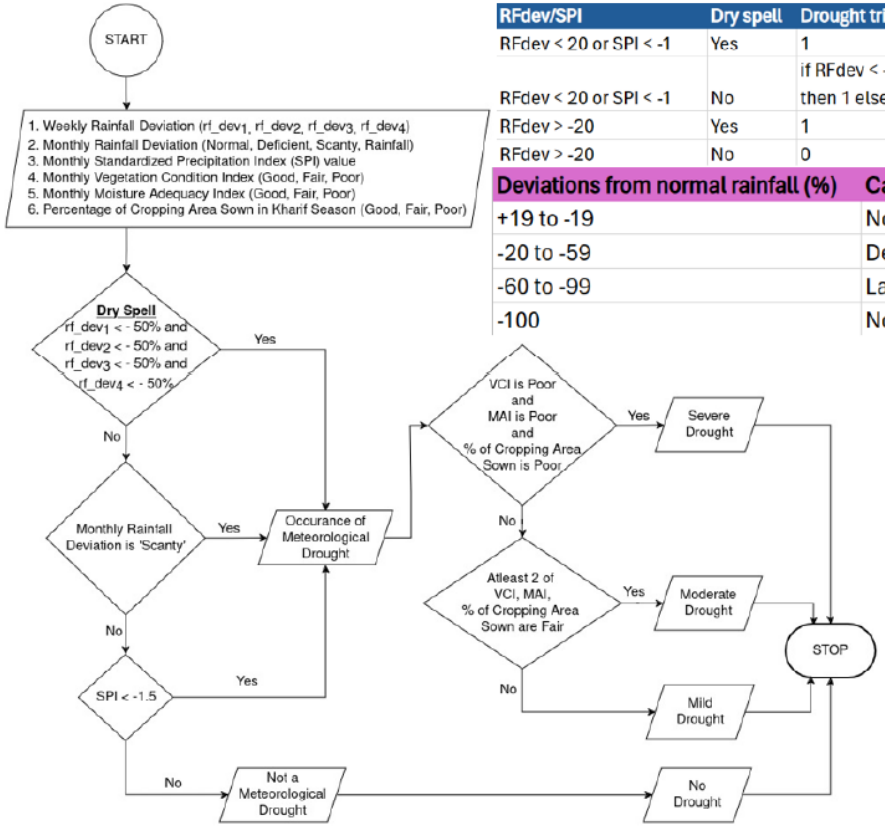
Water Storage Structures

- Existing global products (e.g. Global Surface Water) are effective mainly for large water bodies — they systematically miss structures under ~5,000 m².
- India's national waterbody census (recently released) gives capacity data but is static — no real-time seasonal read.
- **Commons Connect's LULC method can detect water bodies as small as 1,000–5,000 m², and even some sub-1,000 m² bodies, with seasonal (Kharif/Rabi/Zaid) presence.**
- This is fused with the waterbody census to attach capacity information to the remotely-sensed extent.
- Limitation acknowledged directly by the field-testing round: very small farm ponds and traditional structures (ahars, nadis) are still often missed — a real detection gap, not just a resolution nuance.



Drought Severity Assessment

- Based on India's 2016 Manual for Drought Management: combines four index categories — Rainfall, Vegetation, Water, and Crop.
- Government products (e.g. NADAMS) exist but are not publicly available and operate only at district/sub-district scale.
- Commons Connect implements the manual's methodology using operational remote-sensing datasets: CHIRPS rainfall, Landsat-8 NDVI/NDWI, MODIS-derived Moisture Adequacy Index.
- **Output: weekly drought severity (mild/moderate/severe) at micro-watershed granularity — a substantial improvement in spatio-temporal resolution over the state of the art.**



RFdev/SPI	Dry spell	Drought trigger
RFdev < 20 or SPI < -1	Yes	1
RFdev < 20 or SPI < -1	No	if RFdev < -60 or SPI < -1.5 then 1 else 0
RFdev > -20	Yes	1
RFdev > -20	No	0

Deviations from normal rainfall (%)	Category
+19 to -19	Normal
-20 to -59	Deficient
-60 to -99	Large deficient
-100	No rain

SPI value	Category
< -2	Extremely dry
-1.99 to -1.5	Severely dry
-1.49 to -1	Moderately dry
-0.99 to 0	Mildly dry
0 to 0.99	Mildly wet
1 to 1.49	Moderately wet
1.5 to 1.99	Severely wet
> 2	Extremely wet

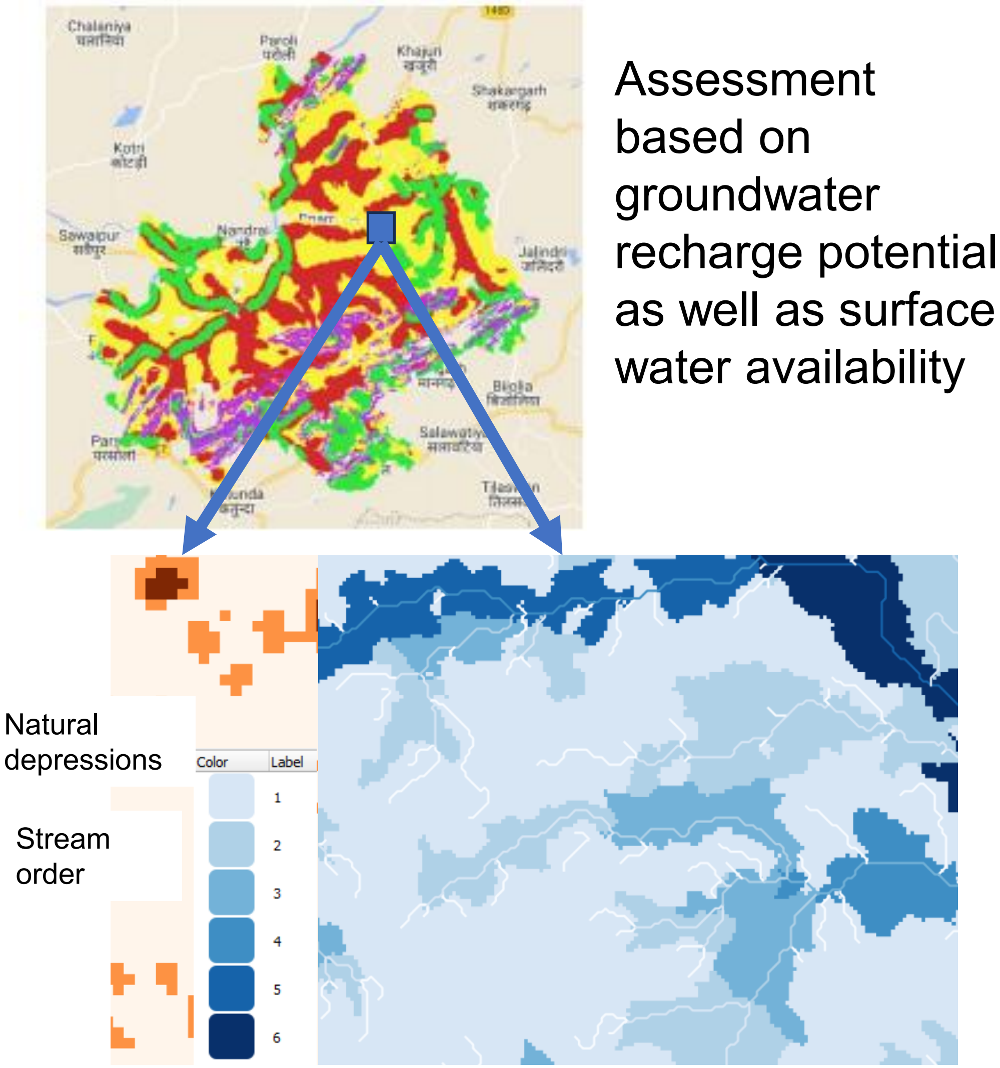
MAI (%)	Category
76-100	No drought
51-75	Mild
26-50	Moderate
0-25	Severe

VCI (%)	Vegetation condition
60-100	Good
40-60	Fair
0-40	Poor

Crop area sown (%)	Category
66-100	Mild
33-66	Moderate
0-33	Severe

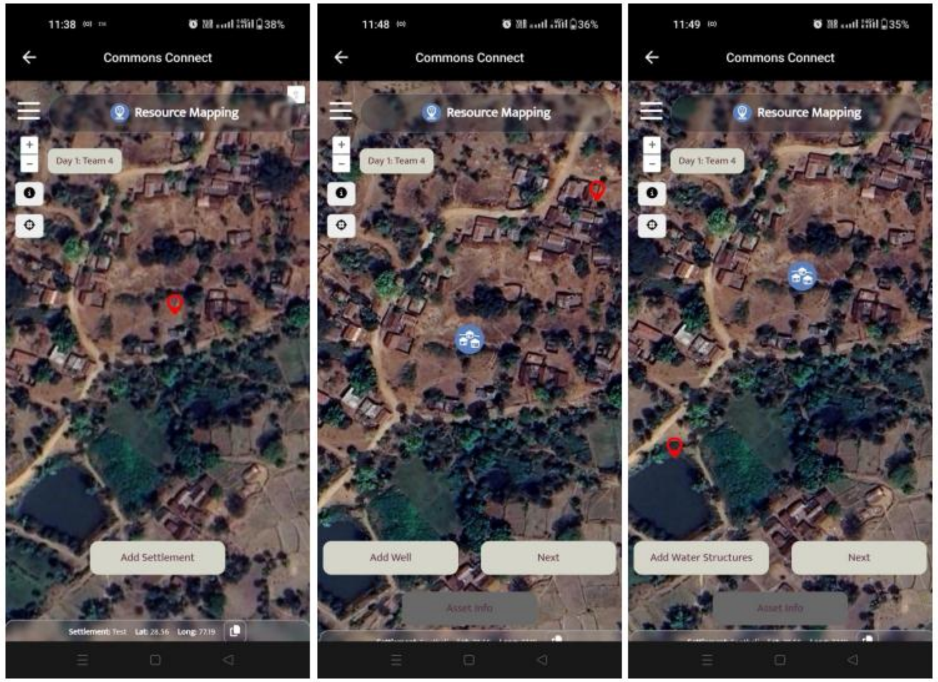
Drainage Lines & Recharge Potential

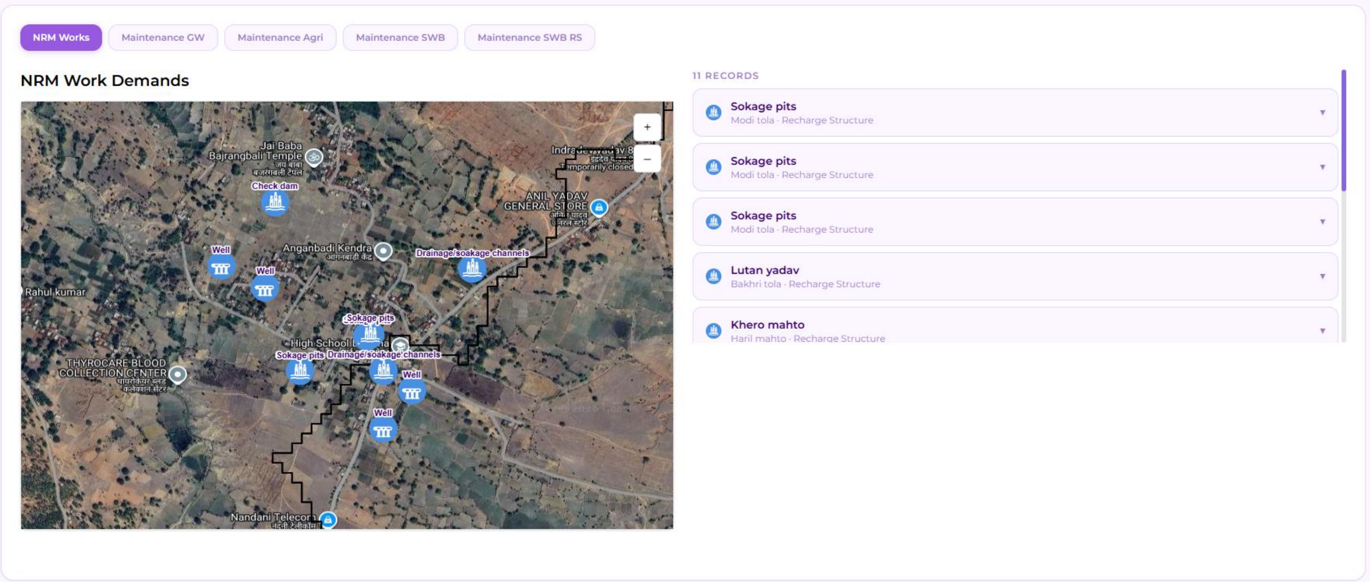
- Drainage lines with stream order: WRIS only publishes stream order 3 and above — too coarse for siting small conservation structures. Commons Connect derives full stream-order hierarchies from DEM flow-accumulation, basin by basin, for the whole country.
- Lower-order streams (1–2) suit loose boulder/rock-filled check-dams; higher orders need concrete structures — a direct design implication of the stream-order label.
- Recharge potential (CLART): a score-based method (FES's algorithm) combining lithology, lineaments, drainage density, and slope into a groundwater-recharge suitability map.
- **Commons Connect re-implements CLART and augments it with its own climatic/landscape indicators to corroborate community demands.**



Resource Mapping in the Field

- Not everything can come from remote sensing — some data must be collected directly with communities.
- Sequential workflow built into the tool: geo-tag a settlement → its wells → its water bodies → sample crop grids (4 ha each) for cropping-intensity estimation of that settlement's land.
- Metadata captured alongside: caste, income level, land ownership, per-settlement asset ownership vs. usage (e.g., a well may be owned by one group but used, for payment, by another).





- **This layer is what makes the equity-mapping work in Topic 4 possible — it is a computing-fundamentals feature with a direct socio-technical payoff downstream.**

FROM DATA TO POLICY

Automated Detailed Project Report (DPR) Generation

- The DPR is the standard document that interfaces village-level demands with the MGNREGA administrative machinery.
- Currently: CSO field staff geo-tag demands manually, then hand-assemble a DPR using separately-obtained geospatial datasets — slow and error-prone.
- **Commons Connect's contribution: community members propose works (type, dimensions, justification, photo) directly on the tool; the DPR — with terrain characteristics and ecological indicators for the relevant micro-watersheds — is generated automatically.**
- CSO feedback in initial testing: automated DPR generation is expected to be a significant time-saver relative to the manual process.

Detailed Info on Wells and their Maintenance demands

Bakhri tola

MWS ID	12_321771
Name of Beneficiary Settlement	Bakhri tola
Type of Well	Open well
Who owns the Well	Privately owned
Beneficiary Name	Lutan yadav
Beneficiary's Father's Name	Nandkishor yadav
Water Availability	June
Households Benefitted	10
Which Caste uses the well?	OBC
Well Usage	Both
Need Maintenance?	Yes
Repair Activities	Repairing_cracks_or_damage_to_the_surface_seal_around_the_well_casing
Latitude	24.44042795
Longitude	86.5486143

Bakhri tola

MWS ID	12_321771
Name of Beneficiary Settlement	Bakhri tola
Type of Well	Open well

PART 3 OF 6

Ethnographic Design & Iteration

Translating a technical approach into something people actually use, through participatory design and real field testing.

Maps to: Commons Connect paper, Section 5 — Field Testing (design-facing findings)

The Field Testing Protocol

- Identical process across all four blocks: community consultation on existing water challenges and MGNREGA use, followed by in-person training of CSO field staff and volunteers on the tool.
- Training covered: marking existing assets and their condition, site-level assessment for new assets, and feedback mechanisms for flagging dataset errors.
- Then, asset-planning exercises were run directly with communities, always in the presence of design researchers.
- **Ethical care built into the protocol: resource-mapping (especially caste/income data collection) was explicitly framed as optional, and the entire exercise was disclosed as field testing — not a live demand-submission process.**
- Notes from all four teams were synthesized afterward to separate location-specific quirks from generalizable design lessons.

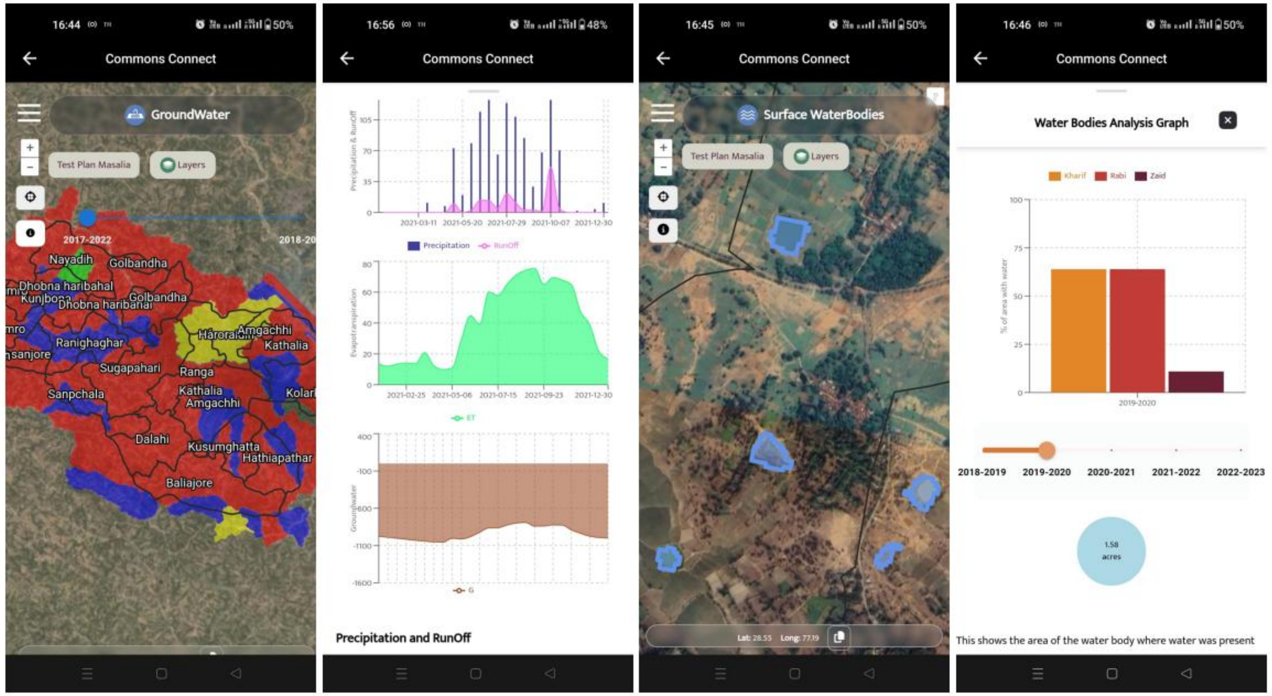
Building Shared Understanding — In Practice

- Standard walkthrough: satellite basemap → zoom to the community's own location → overlay geo-tagged MGNREGA assets → sequentially layer groundwater, surface water, and cropping-intensity screens while scrubbing a time slider.

- Community members actively cross-checked the data against memory: recalling good/bad rainfall years and comparing them to the rainfall time series shown.

- **Where the model's numbers were sometimes off, participants still supplied the right causal story themselves — attributing well-depth drops to expanding double/triple cropping.**

- Net effect: visualizations reliably triggered and sustained discussion, even when not perfectly accurate — the tool worked as a conversation-starter, not just an answer machine.



Data Literacy Was the Real Bottleneck

- Time-series line graphs were meaningfully easier for participants to read than bar charts.
- Sliders (for moving across years) were consistently difficult to operate — a UI pattern that assumes a familiarity many users didn't have.
- More educated field staff and volunteers found textual descriptions of the data easier than the raw charts/maps themselves.
- **Response in progress: template-driven automated text descriptions that narrate anomalies/trends in plain language, plus learning modules linking ecological concepts to the visualizations.**
- Takeaway for design more broadly: sophistication of the underlying model is wasted if the interface doesn't match the audience's existing visual vocabulary.

New Feature Requirements from the Field

Terrain & Site-Level Needs

- Experienced CSO staff wanted upland/midland/lowland terrain classification to match the ridge-to-valley method (e.g. trenches upland, farm ponds lowland).
- Requests for point-level slope (vs. surrounding slope), catchment-area estimates, and drought-sensitivity of runoff at a specific proposed site.
- Response: automated terrain classification via DEM, plus stream-order raster maps not available in round 1.

Skill Tiering & Local Knowledge

- Wide variance in skill across field staff and across CSOs — the design response is a basic/advanced feature split rather than one-size-fits-all complexity.
- **Some siting logic can't come from remote sensing at all: e.g. in Masalia, farm ponds are only useful in midlands (not lowlands) because of local paddy-duration practices — a fact only the community could supply.**
- Land-tenure ambiguity (colonial-era "wasteland" categories) similarly needs community and CSO documentation, not just satellite classification.

Ground-truthing and Offline-First Design

- Systematic validation plan operates at three levels: pixel (e.g. geo-tagged cropping-intensity measurements on a specific farm), object (seasonal water availability recall for a specific water body), and landscape (qualitative community consultation on rainfall/drought/groundwater trends).
- Recall-based data for prior years is known to be unreliable — a real constraint on how far ground-truthing can go retrospectively.
- **Incentive design matters as much as data-collection design: proposed strategy is to piggyback ground-truthing onto activities the user already benefits from (e.g., checking if a structure needs maintenance) rather than as a separate ask.**
- Usability finding with real design consequences: patchy rural connectivity means offline basemaps, passive caching, and local storage with opportunistic sync are now being built as core features, not add-ons.

PART 4 OF 6

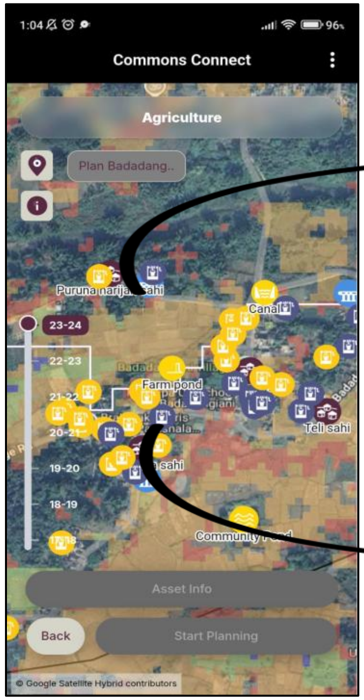
Socio-technical Dynamics at Scale

What happens once something is deployed: how power and inequality get exacerbated or countered, and how intermediaries reshape the system.

Maps to: Commons Connect paper, Sections 5.3 (equity) and 5.4 (MGNREGA integration)

Documenting Unfair Distribution of Assets

- Resource mapping conducted through separate community discussions with each social group residing in different settlements.
- Captured: relative socio-economic status, land holding, population, geo-tagged assets and their ownership/usage, and typical cropping patterns.
- **Key nuance validated in the field: ownership and usage of a resource (e.g. a well) can diverge — one group may own it, another may use it on a paid basis. Groundwater is a shared commons even where the well itself is privately owned.**
- In Bihar, settlements and farmland were not cleanly segregated by group, and plots were fragmented into tiny (5–6m) dismils — the crop-grid sampling method broke down there, forcing reliance on primary discussion data instead of remote sensing.



Settlement with 14 scheduled caste households. Access to only 2 public wells

Land cover: mostly forests

Settlement with 29 OBC households. Access to multiple public and private resources

Land cover: double cropping pattern

Inequity, Made Legible: Piprachak Village

Settlement	Households	Land owned (ha)	Water assets	Cropping intensity
Social group 1	35	25.28	6 (4 ahars, 2 ponds)	2.0 (Kharif + Rabi)
Social group 2	115	50.57	0	1.5 (50% Kharif only)

Group 2 has **3.3× the households** but **zero water assets** and lower cropping intensity — a per-capita gap the raw village-level averages would have hidden entirely.

$$S_i = (1 + r_i) * \frac{\sum_{k=1}^{\text{Total land owned}} CI_{\text{ik}} * a_{ik}}{p_i}$$

$$G = \frac{\sum_{i=1}^{n} \sum_{j=1}^{n} \left|S_i - S_j\right|}{2n^2 \bar{S}_i}$$

Proposed per-capita income-potential score: r_i = water assets, a_{ik} x CI_{ik} = land owned × cropping intensity, p_i = population of group i. Multiplicative form intended to capture how water assets amplify the productivity gains from land — feeding into a Gini coefficient across social groups in a village, to formally flag inequity for MGNREGA allocation decisions.

A CAUTION

The Angul Case: When Legibility Isn't Welcome

- In Angul specifically, the community pushed back against conducting the inequity-mapping exercise at all.
- Their stated reason: caste-based inequities and legacies of different migrant communities settling the area were being managed peacefully — and the exercise risked disturbing that communal harmony.
- **This is an important complication for the course's power-redistribution framing: making inequity legible to outsiders (or to a formal allocation process) is not automatically what an affected community wants, even when the tool's designers believe it serves their interests.**
- The team's response was to make this data-collection module optional/deactivatable based on prior community conversation — treating consent, not just accuracy, as a design requirement.

Whose Interests Does the Data Trail Serve?

Stakeholder	What they want from Commons Connect
CSO field staff / volunteers	Time savings — automated DPR generation vs. manual document assembly
Gram Sabha members	A transparent, trackable demand trail — currently, denials often come with no stated reason
Panchayat members	Ability to prioritize demands using inequity/history data across social groups
MGNREGA technical assistants	Role-based approve/reject access tied to physical site visits
Block programme officers	Budgets automatically attached to demands, to ease cross-panchayat fund review

Same underlying data serves five different institutional interests — none of them identical, and not all necessarily aligned with community empowerment.

Does Transparency Straightforwardly Redistribute Power?

- **Whose power does a given intervention shift, and towards what?**
- Making demands and inequities legible could empower marginalized groups by giving them evidence and a paper trail.
- But legibility is also useful to whoever already controls the tool, the training pipeline, or the Panchayat — the Angul case shows a community anticipating this risk directly.
- **Open questions:**
  - Who decides which data-collection modules are "optional"?
  - Does automated DPR generation shift power toward communities, or toward whichever CSO/official controls the tool's workflow?
  - Can a single technical artifact serve genuinely divergent stakeholder interests without one silently dominating?

PART 5 OF 6

Impact Evaluation

Formally evaluating whether an intervention worked, using causal-inference methods — rather than assuming good intentions imply good outcomes.

Maps to: a 2026 RCT concept note for CommonsConnect, prepared with FES and Inclusion Economics India Centre

Field Testing Is Not Impact Evaluation

- The ICTD paper itself is explicit about its scope: "initial observations from field testing" — usability, dataset accuracy, feature gaps.
- **None of this establishes whether Commons Connect actually changes what gets built, who benefits, or how communities govern shared resources.**
- Good intentions (participatory process, equity mapping, scientific site selection) are hypotheses about impact — not evidence of it.
- A separate 2026 concept note proposes exactly this: a randomized evaluation of Commons Connect + CLART inside a live government planning cycle.

Why Randomize? Treatment vs. Control Panchayats

- The problem with simply comparing GPs that adopt Commons Connect to GPs that don't: adoption itself is not random — motivated CSO staff, engaged leadership, or already-resourced GPs might disproportionately opt in.
- **That selection bias would confound any "the tool worked" conclusion — better outcomes could just reflect who chose to use it.**
- The fix: FES nominates a "long list" of GPs (twice the number they'd like to support this cycle). The research team then randomly assigns half to treatment (tools this cycle) and half to control (status-quo planning, tools offered in a later cycle if desired).
- Random assignment makes treatment and control groups comparable on average — so any difference in outcomes can be credibly attributed to the tool itself, not to who was predisposed to succeed anyway.

Selecting the Study Sample of Gram Panchayats

- Not every GP is an equally informative test case. The concept note proposes prioritizing GPs with:
  - Severe, measurable water and development needs
  - Extensive commons dependence — high shares of grazing/pasture and village-forest land, forest-based livelihoods, recognized Community Forest Rights, reliance on springs or shared surface water
  - Varied caste/tribal composition — needed to observe how inclusively the commons are actually governed, not just whether assets get built
- **This sample-selection logic connects directly back to Topic 4: without caste/tribal variation in the sample, the study couldn't test whether the tool shifts representation at all.**

RESEARCH QUESTIONS

Five Questions the RCT Is Designed to Answer

#	Research question
1	Representation: are SC/ST and women's priorities better reflected in what gets selected and built?
2	Asset selection & placement: are chosen works more climate/water-resilient, and better sited?
3	Gram Sabha quality: does participation and deliberation over shared resources actually change?
4	Community satisfaction: are residents more satisfied with the process and its outcomes?
5	Mechanism: if outcomes shift, is it via new information, transparency, or changed Gram Sabha bargaining?

Question 5 is the one evaluators most often skip — it's what separates "the tool worked" from "here is why it worked," which is what makes a finding actionable elsewhere.

How Impact Would Actually Be Measured

- Administrative data: VB-GRAMG MIS records and planning documents, to assess which assets were approved and prioritized in treatment vs. control GP plans.
- Follow-up survey after the planning cycle, covering village leaders, FES-linked community institutions, and community members across multiple hamlets — capturing participation, collective decision-making over commons, satisfaction, and perceptions of the tools.
- **Baseline survey and existing-asset inventory in study GPs, conducted before planning begins — essential for isolating the treatment effect from pre-existing differences.**
- This two-pronged design (administrative + survey) lets the study triangulate between "what got built" and "how the process felt and functioned" — neither alone would answer the five research questions.

PART 6 OF 6

Operations & Scaling

The institutional and organizational machinery needed to take something from a validated pilot to something that survives at scale.

Maps to: Commons Connect paper, Section 4.4, plus a 2026 field update from Ramgarh, Jharkhand

From Village Demand to Government Portal

- Standard MGNREGA flow: village-level demand → Gram Sabha → Panchayat approval → Block-level fund allocation, interfaced by the Detailed Project Report (DPR).
- Commons Connect's role: automate DPR generation from geo-tagged demands, so what used to be a slow manual assembly step becomes near-instant.
- Where CSOs or volunteers are active, they run PRA-style consultations and feed geo-tagged demands directly into the tool; where State Rural Livelihood Missions are active, Village Livelihood Resource Persons (VLRPs) from Self-Help Group networks can play the same role.
- **As of 2026, Commons Connect can export DPRs directly formatted for upload to Yuktdhara — the Government of India's national geospatial planning portal for MGNREGA — closing the loop from field tool to government system.**

Case: PRADAN in Ramgarh, Jharkhand (2026)

237	9,318	58.5%
villages covered across 6 tehsils	NRM demands collected	of district irrigation is via open wells

- Enabling condition: strong, explicit support from the District Collector — standardized the data-collection approach across departments and hundreds of villages.
- Field workflow: Rozgar Sewaks, Junior Engineers, SRLM staff, and Panchayat functionaries spent ~2 days per village on Commons Connect-based demand capture (asset type, beneficiary gender, geo-tagged location, dimensions).
- Demand pattern: livestock/livelihood infrastructure (cattle, goat, poultry, piggery shelters) was the single largest category (~3,192 demands); water recharge/irrigation works were a close second (~3,309 demands, dominated by wells).
- **Integration step: village-level data consolidated to GP level, converted to KML, and uploaded to Yuktdhara — a manual bridge the team is now systematizing into direct CSV export from Commons Connect.**

What Could Still Go Wrong

- **"Desktop planning" risk: CSOs have long worried that centralized GIS portals like Yuktdhara could let planners determine site selection from an office, sidelining community input — the same risk applies to any tool if the participatory step gets skipped for speed.**
- Funder dependency: current development is backed by a specific set of donors (GIZ, Common Ground, Rainmatter Foundation, Tower Research, and others) — a common vulnerability for public-interest digital infrastructure.
- Administrative dependency: the Ramgarh case succeeded partly because of one District Collector's active backing — a valuable enabler, but also a single point of institutional fragility if leadership changes.
- Data pipeline maintenance: keeping ML models, remote-sensing pipelines, and manual bridge-steps (like the KML/CSV upload to Yuktdhara) running reliably at national scale is a nontrivial, ongoing engineering commitment — not a one-time build.

Cross-Cutting Axes, Threaded Through Every Topic

- Gender: women's SHGs were part of formative consultations (Topic 1); beneficiary gender is now a captured field in Ramgarh demand data (Topic 6).
- Caste: drove exclusion patterns in Mohanpur (Topic 1); shaped the resource-mapping design and the Angul refusal (Topic 4); is a core RCT outcome variable (Topic 5).
- Social capital & marginalization: politically-unconnected and economically marginal households were structurally disadvantaged by MGNREGA's own payment mechanics (Topic 1), and remain the central concern of the equity-mapping module (Topic 4).
- Climate change: motivates the entire water-security framing (Topic 1) and is the explicit target of VB-GRAMG's "climate-resilient" asset selection criterion in the RCT (Topic 5).
- None of these axes appear as a separate module in Commons Connect — they run through the choices made in every other layer of the stack.

Discussion Questions

- The Angul community declined the inequity-mapping exercise. Should Commons Connect ever proceed with it anyway if the research team believes it would help? Who gets to decide?
- Commons Connect's ML models are demonstrably more accurate on some datasets (seasonal water) than others (small waterbody detection, groundwater in alluvium aquifers). How much technical imperfection is acceptable in a tool meant to arbitrate real resource decisions?
- The RCT compares treatment vs. control GPs within one CSO's (FES's) operational area. What would you need to see in the results before recommending the tool for a CSO with a very different working style, like PRADAN in Ramgarh?
- Automated DPR generation clearly saves CSO staff time. Whose time is not being saved by this — and does that matter?

References & Further Reading

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